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The Editor
Monthly Notices of the Royal Astronomical Society

Dear Editor,

I am submitting the manuscript entitled **"HD 338492: A High-Mass Dark Companion Candidate from Gaia DR3 Astrometric–Spectroscopic Binary Data"** for consideration for publication in MNRAS.

This paper presents an archival multi-wavelength analysis of HD 338492, a single-lined spectroscopic binary in the Gaia DR3 non-single-star catalogue whose dynamical properties point to a dormant black hole (BH) companion. The key results are:

- A spectroscopic mass function $f(M) = 2.39\,M_\odot$, which exceeds the Chandrasekhar limit from dynamics alone.
- A minimum companion mass $M_{2,\mathrm{min}} = 6.62\,M_\odot$ (for $M_1 = 4.40\,M_\odot$), well above the neutron-star ceiling.
- No luminous companion detected in any photometric band (optical through mid-infrared), and no X-ray emission.
- Four of six alternative non-BH scenarios excluded on dynamical and photometric grounds; the remaining two (hierarchical triple and stripped helium star) are strongly constrained but require targeted follow-up to close definitively.

What this paper does, and what it does not claim. This work identifies HD 338492 as a strong BH *candidate* based solely on publicly available archival data. We are transparent about its limitations: the stellar characterisation relies on the SIMBAD B9 spectral classification (not independently verified), the SED uses a blackbody approximation, and no independent radial-velocity confirmation exists. We present a sensitivity analysis (Table 4) demonstrating how the BH probability depends on the adopted mass threshold, primary-mass prior, and extinction. Under fiducial assumptions $P(\mathrm{BH}) = 99.7\%$, but this drops to $\sim 89\%$ if the B9 anchor is abandoned entirely — underscoring the importance of follow-up spectroscopy. We explicitly identify the follow-up programme needed to close the remaining gaps: independent multi-epoch radial velocities, a modern spectroscopic classification, and UV observations.

The discovery of Gaia BH1, BH2, and BH3 has demonstrated the power of Gaia astrometric–spectroscopic solutions for finding dormant BHs. These landmark systems were each confirmed with independent RV campaigns. By presenting a carefully vetted archival candidate with a clearly defined follow-up programme, this paper aims to expand the search space for dormant BHs from DR3 and to motivate the observational resources needed for confirmation.

The full analysis pipeline (six reproducible Python scripts), input data, and all intermediate results are publicly available at `https://github.com/jayalabaez/hd338492-black-hole-candidate` and will be archived on Zenodo at proof stage.

No portion of this work has been published or is under review elsewhere. The manuscript is approximately 5 000 words in length with 6 figures and 4 tables.

I suggest the following potential referees with expertise in Gaia compact-object binaries:

- Kareem El-Badry (Caltech) — lead author of Gaia BH1–BH3
- Tomer Shenar (University of Amsterdam) — binary-star spectroscopy and BH candidates
- Tsevi Mazeh (Tel Aviv University) — Gaia spectroscopic binaries

Thank you for considering this submission.

Sincerely,

Joel Ayala-Baez